Weakly Correlated Distillation for Remote Sensing Object Recognition

Wenda Zhao, *Member, IEEE*, Xiangzhu Lv, Haipeng Wang, Yu Liu, *Member, IEEE*, You He, and Huchuan Lu, *Senior Member, IEEE*

***Abstract*— Remote sensing object labels require high specialization, resulting in a limited number of labeled samples. Without large labeled samples to support training, general remote sensing object recognition models have limited accuracy. Addressing this issue, this article proposes a weakly correlated distillation learning framework for remote sensing object recognition with a small number of samples. Benefitting from large-scale natural image datasets, many recognition models achieve superior feature extraction capabilities. Thus, we use them as backbones to build teacher models, and then fine-tune the teacher models with a small-scale remote sensing dataset. However, due to the limited number of remote sensing samples, the teacher models may produce noisy features that reduce the performance of the student model. Therefore, we propose a weakly correlated distillation method that selects the weakly correlated features from teacher models to distill the student. Since the weakly correlated features contain different noise distributions that can be mutually suppressed, thereby improving the performance of the student. Extensive experiments on three widely-used datasets of DOTA, HRRSD, and NWPU VHR-10 demonstrate the superior performance of our method compared with the state of the arts (SOTA). Code is available at: https://github.com/wdzhao123/WCD.**

***Index Terms*— Limited samples, remote sensing object recognition, weakly correlated distillation.**

I. Introduction

REMOTE sensing object recognition based on deep learning [1], [2], [3], [4], [5], [6] has been a significant research topic. General natural image datasets, such as ImageNet [7] and CIFAR10 [8], have a large number of samples, which are able to train models to obtain high object recognition accuracy. However, on the one hand, some remote sensing images are difficult to obtain due to scarce objects, such as aircraft carriers. On the other hand, the acquirement of remote sensing labels needs professional manual annotation, which is expensive and laborious. Therefore, when deep learning models relying on a large number of samples for training are applied to limited remote sensing images, their object recognition accuracy and generalization performance will decrease. For example, in Fig. 1, trustworthy long-tailed classification (TLC) [9] is trained on DOTA training dataset and achieves accuracy of 79% on DOTA testing dataset, but its testing performance decreases by 35.5%, 23.8% on HRRSD dataset and NWPU VHR-10 dataset, respectively.

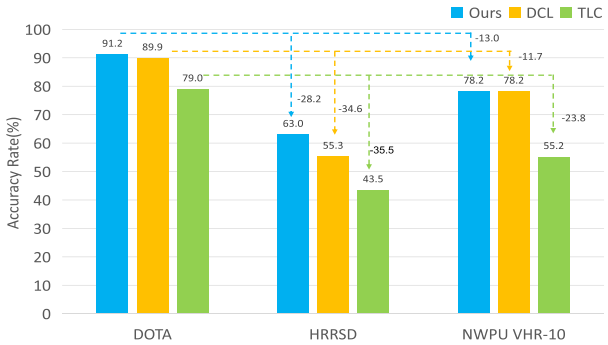


Fig. 1. Performance comparison among ours, DCL [10] and TLC [9] on DOTA dataset [11], HRRSD dataset [12] and NWPU VHR-10 dataset [13]. These methods are trained on the DOTA training dataset, and tested on the testing datasets of HRRSD and NWPU VHR-10, respectively. Compared with the state-of-the-art (SOTA) DCL and TLC, our method achieves superior performance on the DOTA dataset and better generalization on the HRRSD dataset and the NWPU VHR-10 dataset.

Generally, there are three solutions to address the small-sample problem: meta-learning, data augmentation, and transfer learning. Meta-learning [14], [15], [16] aims to equip deep networks with the ability to learn "how to learn," thereby allowing them to quickly adapt to new classification tasks with a small number of samples. However, meta-learning training process is too complicated to train. Data augmentation [17], [18], [19] increases the number of samples by some methods, such as image style conversion. However, the qualities of the augmented images are uneven, and thus the training result is subjected to the quality of the composite images. Transfer learning [20], [21], [22] is to train a backbone with large samples and then fine-tune the backbone with small samples. However, traditional transfer learning tends to introduce noise because of insufficient samples.

In this article, we propose a weakly correlated distillation learning to relieve the small-sample remote sensing object

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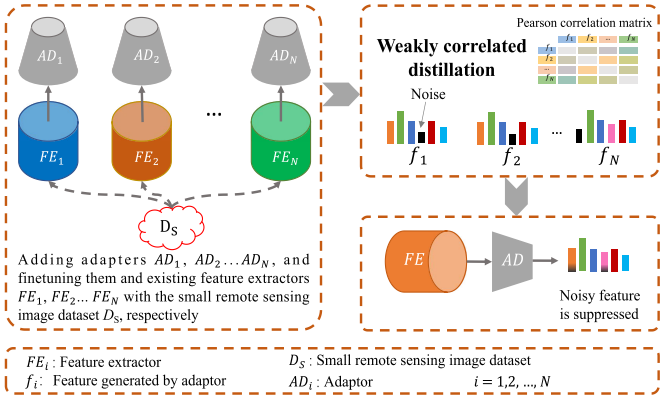


Fig. 2. Motivation of the proposed weakly correlated distillation learning. The training process is divided into two stages. In the first stage, a small remote sensing dataset is used to fine-tune feature extractors and adaptors of teacher models. In the second stage, the Pearson correlation matrix is utilized to find the weakly correlated features, and then distill the student that noisy feature is suppressed.

recognition problem. Considering many recognition models benefited from large-scale natural image datasets achieve superior feature extraction capabilities, we can take advantage of their feature extraction capabilities by distillation learning, as shown in Fig. 2. However, there is a main problem that small remote sensing samples will make teacher models produce noise during the distillation process, thereby reducing the performance of the student model. Addressing this problem, we propose the weakly correlated distillation that selects the weakly correlated features from teacher models to distill the student. Since the weakly correlated features contain different noise distributions that can be mutually suppressed, and then improve the distillation effect using small remote sensing samples.

Specifically, given the small remote sensing dataset D_S, we select N different existing recognition networks as the teacher models to finetune. Since the original models are only applicable to the natural images, adaptors $\{\mathrm{AD}_1, \mathrm{AD}_2, \ldots, \mathrm{AD}_N\}$ are added to the teacher models that adapt to the remote sensing images. Then, we finetune the teacher models with D_S. However, a small number of remote sensing samples is not enough to train the models adequately, thus these models are prone to generate plenty of noisy features. Therefore, we calculate the Pearson correlation matrix between the features of each teacher model. Through the Pearson correlation matrix, the weakly correlated teacher models with the different features are distilled the student model, so as to offset the influence of noise. Hence, the student model can get high-precision prediction results under the small remote sensing dataset.

This article's contributions can be summarized as follows.

1) We explore the small-sample remote sensing object recognition, and propose a weakly correlated distillation learning method to address this.
2) We measure the feature relationship of teacher models and select the weakly correlated features to distill student model that offsets the impact of noisy features from different teacher models.
3) Extensive experiments on three widely used datasets of DOTA, HRRSD, and NWPU VHR-10 demonstrate the superior performance of our method.

The remainder is organized as follows. First, related work is discussed in Section II. Second, weakly correlated distillation learning framework is introduced in Section III. Third, extensive experiments are built in Section IV. Finally, our work is summarized in Section V.

II. Related Work

A. Object Recognition

Nowadays, object recognition is still a hot topic in the field of computer vision, and many recent works focus on improving the feature extraction ability of object recognition. ResNet [23], VGG [24], DenseNet [25] et al. are the most typical and basic fully-supervised classification networks, which are still widely used as baselines today. Nuriel et al. [26] propose a permuted adaptive instance normalization to reduce the representation of global statistics in the hidden layers of image classifiers. Mackowiak et al. [27] develop a generative classifier architecture, which is able to deal with more complex scenes. Li et al. [9] identify hard samples in a multiexpert framework by jointly conducting classification and uncertainty estimation for long-tailed classification. Yang et al. [28] propose a novel graph-based object representation as an auxiliary feature regularization and a pose-insensitive constraint on shallow-to-deep sub-network for fine-grained object classification.

In recent years, many object recognition methods in remote sensing field have also been proposed. Zheng et al. [29] proposed a task-specific classifier framework to enhance domain adaptation ability for high spatial resolution remote sensing classification. Lv et al. [30] propose a spatial-channel feature preserving vision transformer model to settle the limited stacked receptive field issue. Xu et al. [31] propose a lie group regional influence network to extract more distinguishing ability and effective features of remote sensing images. Zhao et al. [1] propose a feature balance strategy to address fine-grained remote sensing object classification. Zhao et al. [32] implement style-content metric learning for multidomain remote sensing object recognition.

However, the existing methods need a considerable amount of training samples, and thus suffer from performance degradation when they are trained with limited samples. In contrast, we propose a weakly correlated distillation learning method, which can achieve high accuracy with only a small amount of training samples.

B. Small-Amount-of-Sample Solution

Small-amount-of-sample learning aims to train the model with fewer samples, but to achieve higher accuracy. Various methods have been proposed, such as meta-learning [33], [34], [35], transfer learning [2], [36], [37], [38], and data augmentation [10], [17], [18]. Lee et al. [35] propose Task Discrepancy Maximization to locate class-level identification areas by highlighting channels that encode different information about the class. Chen et al. [33] employ comparison, squeeze, and

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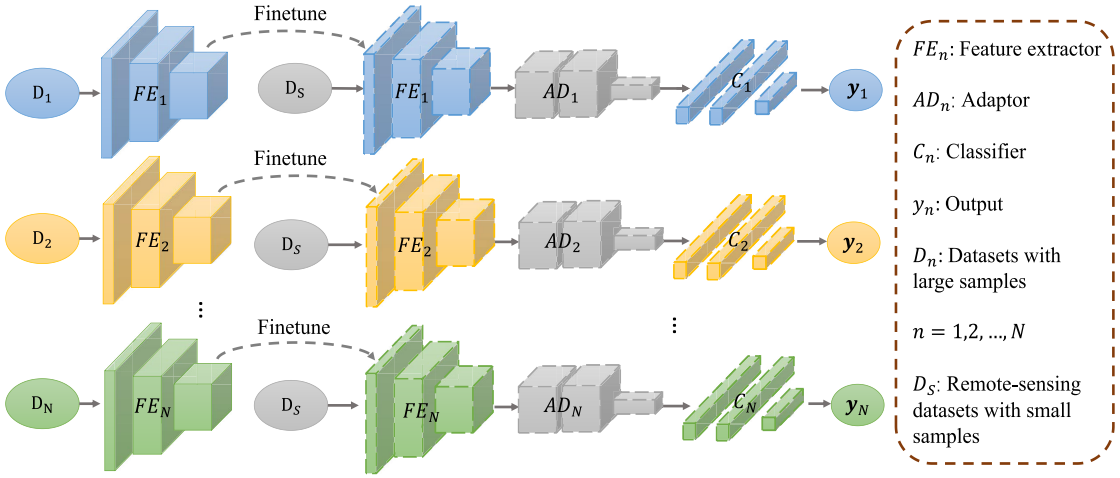


Fig. 3. Illustration of multiple teacher generation. Given N teacher models pretrained with large natural image datasets $\{D_1, D_2, \ldots, D_N\}$, we use their feature extractor FE_n, and add adapter AD_n and classifier C_n. Then, we finetune these new teacher models (dashed border) with a small remote sensing dataset D_S, so as to they are suitable for remote sensing object recognition.

calibration modules to explore the class-level knowledge in the few-shot classification task. Dong et al. [36] propose a simple but effective baseline for unforeseen adversarial few shot image classification task by directly learning generalized representations without tedious task sampling. Xiao et al. [38] design cross-domain spatial structural consistency loss and disturbance correlation consistency loss to calibrate the target generative models. Cubuk et al. [17] search for the best strategy automatically to achieve data enhancement. Zhang et al. [18] exploit class-conditional training and supervised generative adversarial network to deal with limited-labeled-sample problem.

Since the number of remote sensing samples is small, traditional transfer learning will introduce feature noise, thereby limiting the accuracy of the student model. Therefore, we select the weakly correlated features from teacher models to distill the student, and thus different noise distributions can be mutually suppressed.

C. Knowledge Distillation Learning

Various knowledge distillation methods have been proposed, such as learning from multiple teachers methods [39], [40], [41], [42], [43], teacher assistant methods [44], [45], [46], self-distillation [47], [48], [49], [50]. Learning from multiple teachers is to transfer the knowledge of multiple teachers to a single student model so as to improve the performance of the student model. Fukuda et al. [42] improve the performance of a single student model by randomly selecting the soft target of a teacher model. Jiang et al. [40] improve the student model by providing both long-term teachers with stable signals and short-term teachers with high-quality training updates. Teacher assistant serves as an intermediate bridge to compensate for the gap between teachers and students. Mirzadeh et al. [46] use a medium-sized network to act as a teacher's assistant. Self-distillation means that a single network is used as both a teacher and a student model, so that a single network model can improve its performance through knowledge distillation in the process of self-learning. Zhang et al. [50] use the features of the deep network to guide the learning of the shallow network to achieve self-distillation.

However, one-teacher distillation can hardly suppress noise in the teacher model when training with small samples. In multiteacher distillation, the selection of teacher models is random or average, failing to find the most suitable combination of teacher models to distill. Therefore, our method selects weakly correlated teacher models through the Pearson correlation matrix. Since the teacher models with the least correlation degree have the largest noise difference, the influence of noise will be offset by being mutually suppressed, so that achieves higher accuracy of the student model.

III. Weakly Correlated Distillation Learning

A. Motivation and Framework

Small remote sensing image samples may lead to low accuracy of remote sensing objection recognition. Knowledge distillation is an effective way to solve this problem, in which the teacher model acquires information from a large number of natural image samples and then distills it into a student model trained with small remote sensing image samples. Generally, the traditional one teacher distillation model only learns from a single teacher. No matter whether the teacher's information is beneficial or useless, the students all accept it. The multiteacher distillation model provides multiple information flows to interpret the task of the student model and thus can produce better performance. However, since the number of remote sensing image samples is small, the teacher model will generate features with noise that will make the student model's performance decline during the distillation process. Therefore, how to choose appropriate teacher models from multiteacher models to suppress noise is a crucial issue.

Based on the above discussion, we propose a weakly correlated distillation model that aims at selecting the weakly correlated features of multiple teacher models to distill the student. Our method is divided into two stages: multiple teacher generation and weakly correlated distillation, as shown in Figs. 3 and 4. Details are described as follows.

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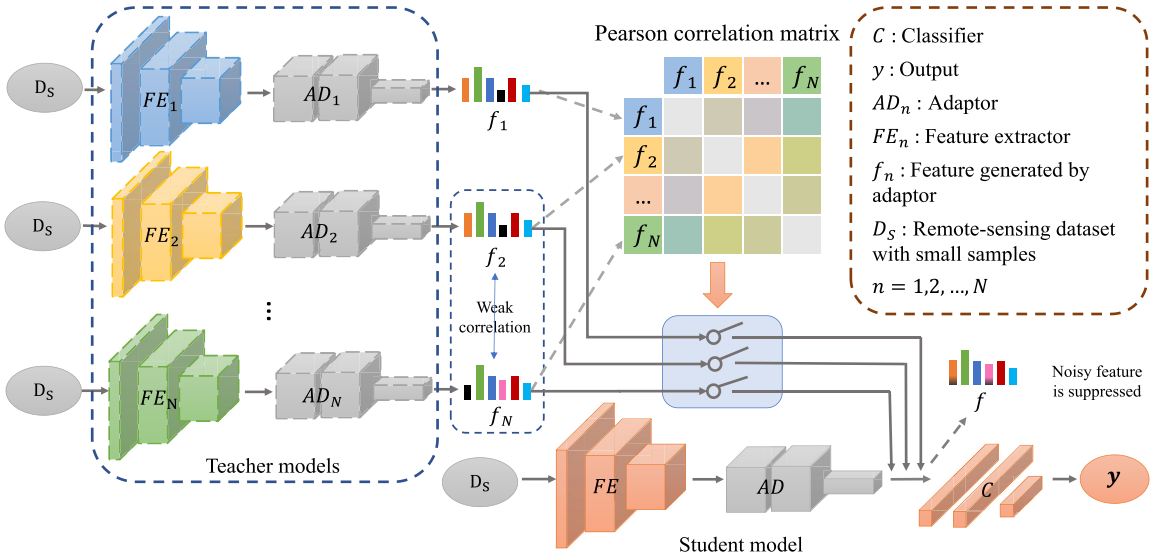


Fig. 4. Illustration of weakly correlated distillation. The remote sensing sample is fed to multiple teacher models, and then multiple features are generated. After that, the Pearson correlation matrix between the features of the teacher models is calculated. Lastly, two features with the weakest correlation coefficient are selected to distill the student model.

B. Multiple Teacher Generation

Due to sufficient natural image samples, we can get high-accuracy teacher models for natural image object recognition. Sequentially, we fine-tune the teacher models to improve their feature extraction capabilities with the small remote sensing dataset, as shown in Fig. 3. Specifically, given N teacher models pretrained with large natural image datasets $\{D_1, D_2, \ldots, D_N\}$, we fine-tune the teacher models with small scale remote-sensing dataset $D_S = \{(d_1, g_1), \ldots, (d_N, g_N)\}$, where $d_n (n = 1, 2, \ldots, N)$ represents the nth labeled sample with spatial dimension of $H \times W$, $g_n (n = 1, 2, \ldots, N)$ is the label of d_n, and $g_n \in R^{C\times 1}$ has C categories. In addition, in order to adapt the teacher models to the remote sensing object recognition, we add an adaptor AD_n between the feature extractor FE_n and the classifier C_n of each teacher model.

Cross-entropy loss L_n is utilized as the supervision of the nth teacher model, which is defined as

$$L_n = -\sum_{i=1}^{M} y_i^n \log g_i^n \tag{1}$$

where M denotes the total number of samples, y_i^n signifies the prediction output of the ith sample, and g_i^n indicates the ground-truth of the ith sample.

C. Weakly Correlated Distillation

Since teacher models fine-tuned by small scale remote-sensing dataset tend to generate noise, traditional multiteacher distillation that uses random distillation to select the teacher models will cause the defect of inaccurate results. Therefore, we design the weakly correlated distillation method, as shown in Fig. 4.

We observe that noise distributions are different produced by multiple teacher models, and we can adopt feature correlation to measure them. As shown in Fig. 5, two strongly correlated features are selected to distill the student model.

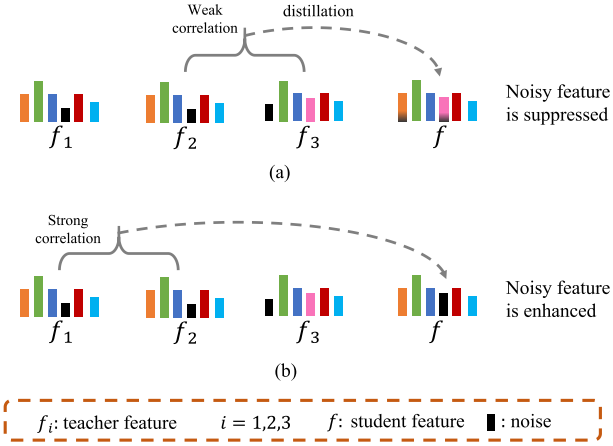


Fig. 5. (a) Two weakly correlated features f_2 and f_3 are used to distill the student model, and the noisy feature is suppressed. (b) Two strongly correlated features f_1 and f_2 are used to distill the student model, and the noisy feature is enhanced.

The noise produced by the student will be enhanced due to the superposition of noise at the same location. On the contrary, two weakly correlated features are selected to distill the student model. The noise produced by the student will be suppressed due to the average noise at different locations. Therefore, we utilize the Pearson correlation matrix to measure the feature correlation, and then dynamically select two weakly correlated features generated by teacher models to distill the student model. Besides, due to the limited samples, the object feature extracted by the teacher models may be incomplete beyond the noise. Given an example using a ship, it is possible that one teacher model fails to extract its prow feature while another teacher model fails to extract its stern feature. Fortunately, we choose weakly correlated features generated by teacher models to distill the student, thereby utilizing the complementarity of these diverse features to improve performance.

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TABLE I

NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CATEGORY OF THE DOTA DATASET AND THE NUMBER OF TEST SAMPLES FOR EACH CATEGORY OF HRRSD DATASET AND NWPU VHR-10 DATASET

Name	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
Category	Baseball-diamond	Basketball-court	Bridge	Ground-track-field	Harbor	Airplane	Ship	Vehicle	Storage-tank	Tennis-court
DOTA-Train	42	52	106	33	298	399	281	261	251	118
DOTA-Test	214	132	464	144	2090	2531	8960	5438	2888	760
HRRSD-Test	1001	832	1090	795	938	1217	935	1178	1095	1011
NWPU VHR-10-Test	390	159	124	163	224	757	302	598	655	524

TABLE II

EFFECTIVENESS STUDY OF NOISE SUPPRESSION. THE TEACHER MODEL AND THE CORRESPONDING STUDENT MODEL ARE TRAINED ON DOTA TRAINING DATASET, AND TESTED ON DOTA TEST DATASET

Method		Acc.	Pre.	Rec.
VGG16	Teacher	89.1	77.9	**81.2**
	Student	**90.7**	**82.4**	80.4
ResNet50	Teacher	88.2	74.3	76.9
	Student	**90.9**	**79.7**	**82.0**
DenseNet121	Teacher	87.8	72.7	77.0
	Student	**91.2**	**81.7**	**85.4**

Specifically, we extract the intermediate feature after the adaptor AD_n as the soft label for distillation. Let $X_S = [x_1, x_2, \ldots, x_S]$ represent a batch of samples, where S is the batch size. X_S is fed into N teacher models to get N features $F_N = \{f_1, f_2, \ldots, f_N\}$, where $f_n \in \mathbb{R}^D$ and D is the dimension of the feature. Then, we calculate their Pearson correlation matrix $A \in \mathbb{R}^{N\times N}$ as follows:

$$A_{ij} = \frac{\sum_{d=1}^{D}\left(f_i^d - \bar{f}_i\right)\left(f_j^d - \bar{f}_j\right)}{\sqrt{\sum_{d=1}^{D}\left(f_i^d - \bar{f}_i\right)^2}\sqrt{\sum_{d=1}^{D}\left(f_j^d - \bar{f}_j\right)^2}} \tag{2}$$

where $f_i = \{f_i^1, f_i^2, \ldots, f_i^D\}$, $f_j = \{f_j^1, f_j^2, \ldots, f_j^D\}$ are feature vectors, $\bar{f}_i$, $\bar{f}_j$ are the average of the feature vectors f_i, f_j, i.e., $\bar{f}_i = (1/D)\sum_{d=1}^{D} f_i^d$, $\bar{f}_j = (1/D)\sum_{d=1}^{D} f_j^d$.

We choose two features $\{f_{mi}, f_{mj}\}$ with the weakest correlation coefficient for the sample m to distill the student model (see Fig. 4), and the distillation loss L_{dis} is described as follows:

$$L_{\mathrm{dis}} = \sum_{m=1}^{M}(f_m - f_{mi})^2 + \sum_{m=1}^{M}\left(f_m - f_{mj}\right)^2 \tag{3}$$

where f_m is the feature of the mth sample generated by the student model, and M is the total number of samples.

Besides, cross-entropy loss L_s is utilized to supervise the student model, which is defined as

$$L_s = -\sum_{m=1}^{M} y_m^s \log g_m^s \tag{4}$$

Algorithm 1 Optimization Process of Our Framework

Input: Small scale remote-sensing dataset D_S.
Stage 1: Training the teacher models with D_S.
1. Optimizing the teacher models using Equation (1).
Stage 2: Fixing teacher models and training the student model.
2. Selecting the weakly correlated features generated by teacher models.
3. Optimizing the student model using Equation (5).
Output: Parameters of the student model.

where y_m^s is the output of the student model for the mth sample and g_m^s is the corresponding ground-truth.

Thus, the overall loss of the student model is formulated as

$$L_{\mathrm{all}} = L_s + \beta L_{\mathrm{dis}} \tag{5}$$

where β is a hyper-parameter to adjust the distillation.

D. Optimizing

1) Structure: Considering the efficiency, we set the number of teacher models N to 3, i.e., ResNet50 [23], VGG16 [24], DenseNet121 [25] are adopted as the backbones. We choose the first four blocks of ResNet50, the first five blocks of VGG16, and the first four blocks of DenseNet121 as feature extractors of the teacher models. The adaptors are added, respectively, which consist of two $3 \times 3 \times 512$ convolution layers and a global average pooling. Besides, three fully connected layers are added as the classifier, whose sizes are 512, 512, and 10, respectively. The structure of the student model is the same as the corresponding teacher model.

The optimization process of our framework consists of two stages, as shown in Algorithm 1. The first stage is to fine-tune teacher models with small-scale remote-sensing dataset D_S, and L_n is used as the loss function. The second stage is to fix on the teacher models and train the student model. First, the weakly correlated features generated by teacher models are selected. Second, the student model is optimized using the loss L_{all}.

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TABLE III

GENERALIZATION STUDY OF NOISY FEATURE SUPPRESSION. THE TEACHER MODEL AND THE CORRESPONDING STUDENT MODEL ARE TRAINED ON DOTA TRAINING DATASET, AND TESTED ON HRRSD DATASET AND NWPU VHR-10 DATASET, RESPECTIVELY

Dataset	VGG16						ResNet50						DenseNet121					
	Acc.		Pre.		Rec.		Acc.		Pre.		Rec.		Acc.		Pre.		Rec.	
	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student	Teacher	Student
HRRSD	**56.7**	49.9	**67.3**	62.2	**56.1**	49.7	56.0	**60.1**	61.3	**69.3**	55.7	**60.2**	53.9	**63.0**	63.3	**71.2**	53.8	**63.7**
NWPU VHR-10	77.5	**77.8**	69.9	**75.2**	70.1	**72.1**	68.8	**75.5**	61.8	**70.8**	62.5	**71.3**	72.1	**78.2**	64.9	**73.4**	65.3	**74.3**

IV. EXPERIMENT

A. Configuration

1) Dataset: Three remote sensing classification datasets are used: DOTA dataset [11] collected from Google Earth, GF-2 and JL-1 satellite, HRRSD dataset [12] collected from Google Earth and Baidu Map and NWPU VHR-10 dataset [13] collected from Google Earth and Vaihingen. As shown in Table I, 1841 training samples from DOTA are used to train the model, and 23 621, 10 092, and 3896 test samples from DOTA, HRRSD, and NWPU VHR-10 are used for testing the model, respectively. To verify the object recognition accuracy and generalization of the proposed method, ten common categories shared by the three datasets are selected, which are baseball-diamond, basketball-court, bridge, ground-track-field, harbor, airplane, ship, vehicle, storage-tank, and tennis-court. A large number of samples can achieve good performance of object recognition, e.g., 97 500 images are built in [51]. In contrast, we attempt to explore the small-sample remote sensing object recognition, and the number of our training samples is 1841.

2) Implementation: The experiment is conducted on a GTX 2080Ti GPU using PyTorch. The optimizer is Adam with the learning rate of $1e^{-3}$. The batch size is set to 8, and input images are 224 $\times$ 224 pixels. Training is divided into two stages: training the teacher models and training the student model. The training epochs of the two stages are both 1000. Hyper-parameter β is taken to 0.5.

3) Metrics: We use three metrics to verify the performance of our method, namely accuracy (Acc.), precision (Pre.) and recall (Rec.). Acc. is the ratio of the correctly predicted samples to all predicted samples, whose formula is as follows:

$$\text{Acc.} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{6}$$

where TP is the number of positive ground-truth samples that are predicted to be positive samples, FN is the number of positive ground-truth samples that are predicted to be negative samples, TN is the number of negative ground-truth samples that are predicted to be negative samples and FP is the number of negative ground-truth samples that are predicted to be positive samples. Pre. is the ratio of the correctly predicted positive samples to all predicted positive samples, whose formula is as follows:

$$\text{Pre.} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{7}$$

TABLE IV

EFFECTIVENESS STUDY OF WEAKLY CORRELATED DISTILLATION BY COMPARING DIFFERENT DISTILLATION METHODS. RESNET50 IS SELECTED AS THE STUDENT MODEL

Teacher Model	Acc.	Pre.	Rec.
VGG16+DenseNet121	89.9	**81.5**	81.3
ResNet50+DenseNet121	89.0	76.2	79.0
VGG16+ResNet50	90.1	81.1	**82.6**
VGG16+ResNet50+DenseNet121	89.6	78.7	80.1
Strongly correlated distillation	89.3	76.5	81.0
Ours	**90.9**	79.7	82.0

TABLE V

STUDY ON THE STUDENT MODEL WITH ONE AND TWO ORDERS OF MAGNITUDE FEWER PARAMETERS

Model	Parameters	Acc.	Pre.	Rec.
Teacher	$3.58 * 10^7$	88.2	74.3	76.9
Raw student	$3.58 * 10^7$	90.9	79.7	82.0
1 order of magnitude lower	$6.74 * 10^6$	90.2	80.5	83.3
2 orders of magnitude lower	$8.06 * 10^5$	90.0	80.4	79.9

where C is the number of categories. Rec. is the ratio of correctly predicted positive samples to all true positive samples, whose formula is as follows:

$$\text{Rec.} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}. \tag{8}$$

B. Ablation Study

1) Effectiveness of Noisy Feature Suppression: The teacher model will produce noisy features due to the training by the small-scale remote sensing dataset, and this limits the performance of the teacher model. Therefore, we select appropriate teacher models to implement the weakly correlated distillation that suppresses the noise, thereby improving the recognition accuracy. To verify this, three backbones ResNet50, VGG16, and DenseNet121 are used to implement the teacher model and the corresponding student model. We compare the results of the student model with the corresponding teacher model, as shown in Table II. The results of the student models are superior to the corresponding teacher models on the DOTA dataset. In addition, we verify the generalization of the proposed method by testing it on the HRRSD and NWPU VHR-10 datasets. As can be seen from Table III, the student

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TABLE VI

RESULTS OF OUR METHOD (OURS) AND SOTA METHODS ON DOTA DATASET. ACC., PRE., REC. AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	Pre.	Rec.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	70.9	66.3	69.1	71.5	78.0	26.5	**66.7**	48.2	94.2	85.9	30.5	**97.8**	**91.4**
TLC	79.0	65.7	60.4	34.6	18.9	51.7	18.8	79.6	85.5	77.2	80.2	83.2	74.5
Sagnet	81.5	64.2	67.1	69.2	45.5	33.2	36.1	76.0	84.3	83.5	83.0	89.4	70.9
RIDE	82.4	69.3	70.5	44.9	77.3	68.5	13.2	**95.5**	92.8	82.5	78.3	83.4	68.4
DCL	89.9	81.1	84.7	89.7	**83.3**	**85.3**	48.6	84.5	95.7	86.6	**95.1**	92.9	85.5
Ours	**91.2**	**81.7**	**85.4**	**90.2**	77.3	71.1	66.0	89.9	**97.1**	**91.2**	90.7	94.7	86.2

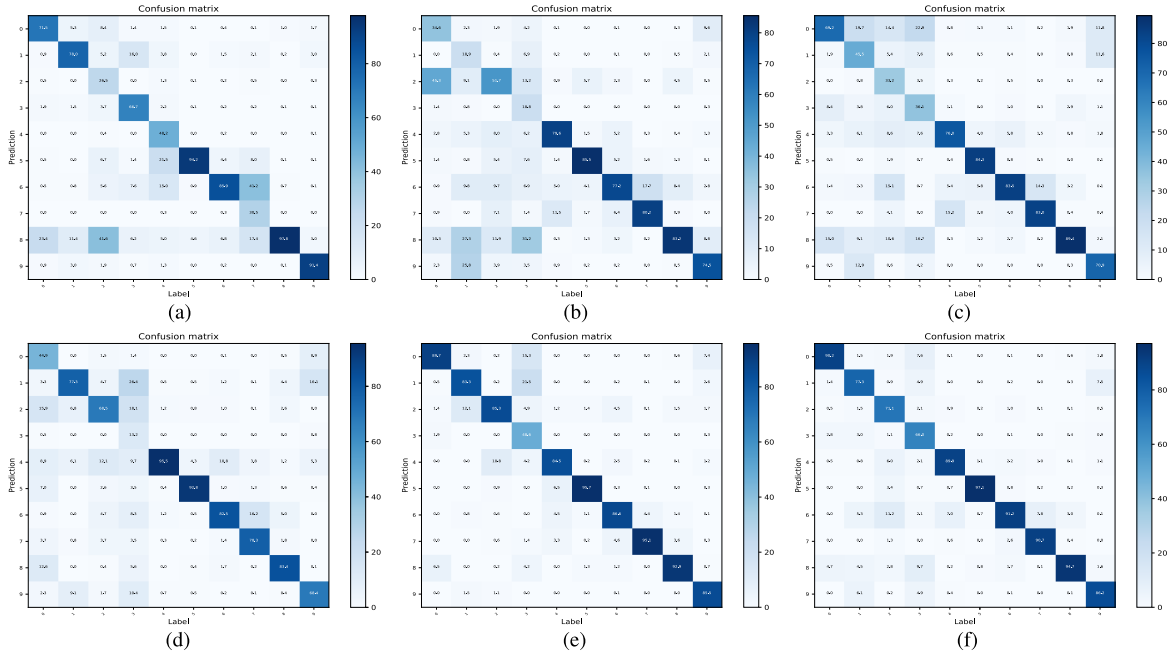


Fig. 6. Confusion matrices of different methods on DOTA. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. (a) PMG. (b) TLC. (c) Sagnet. (d) RIDE. (e) DCL. (f) Ours.

TABLE VII

RESULTS OF OUR METHOD (OURS) AND SOTA METHODS ON HRRSD DATASET. ACC., PRE., REC. AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	Pre.	Rec.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	54.3	62.1	53.9	**88.4**	34.3	**13.7**	43.4	77.7	80.9	15.4	2.0	94.5	89.3
TLC	43.5	47.3	43.1	44.6	25.2	0.7	31.6	94.1	82.0	18.7	7.7	66.7	59.6
Sagnet	46.6	48.1	45.9	48.6	33.4	0.6	34.0	87.5	87.8	16.8	13.2	83.7	53.9
RIDE	49.9	55.6	49.7	24.2	**78.5**	11.0	24.9	95.4	**92.5**	21.4	13.4	84.2	51.1
DCL	55.3	**72.0**	54.4	47.7	29.1	6.1	36.6	**99.4**	86.5	23.9	**27.2**	**97.7**	**90.2**
Ours	**63.0**	71.2	**63.7**	73.2	62.6	8.2	**69.1**	92.9	92.1	**54.0**	14.3	97.2	73.4

models achieve better accuracy than the teacher models. Especially, Bayraktar et al. [52] filters unrelated areas in the image to improve object detection. From another perspective, our method suppresses feature noises to obtain good performance, as shown in Tables III and II.

2) Effectiveness of Weakly Correlated Distillation: Weakly correlated distillation selects the weakly correlated features generated by teacher models to distill to the student model, where diverse features are adopted to improve the performance of the student. To verify the effectiveness of the weakly correlated distillation, different combinations of teacher models and strongly correlated distillation are compared, as shown in Table IV. Our method achieves the best performance compared to the other methods in terms of Acc.

3) Effect of Parameters: We study on the effect of parameters by adjusting the parameters of the student model to one and two orders of magnitude lower than the original. Specifically, we reduce the number of convolution blocks in

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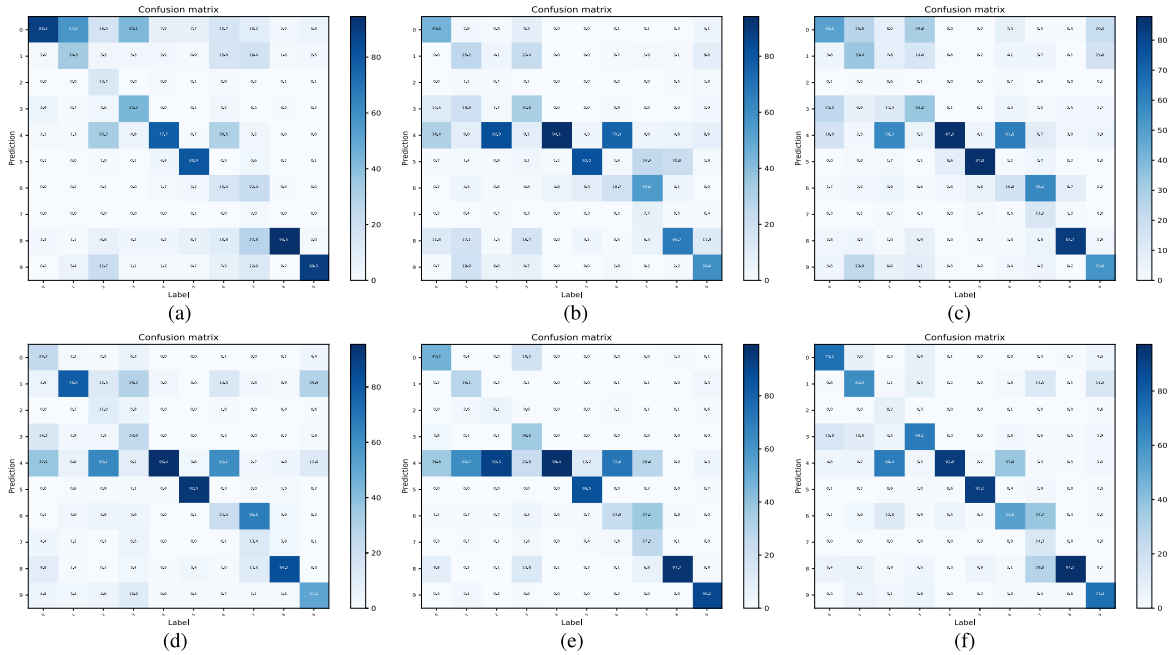


Fig. 7. Confusion matrices of different methods on HRRSD. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. (a) PMG. (b) TLC. (c) Sagnet. (d) RIDE. (e) DCL. (f) Ours.

TABLE VIII

RESULTS OF OUR METHOD (OURS) AND SOTA METHODS ON NWPU VHR-10 DATASET. ACC., PRE., REC. AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	Pre.	Rec.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	73.6	71.6	71.0	94.6	**98.1**	**15.3**	**98.2**	64.7	**99.5**	57.3	10.7	98.8	72.9
TLC	55.2	52.8	48.0	53.8	10.1	0.8	42.9	79.0	95.1	65.9	19.1	44.6	68.7
Sagnet	55.2	48.4	49.3	41.3	28.9	0.0	55.8	95.1	91.1	49.0	19.7	74.2	38.0
RIDE	62.6	58.4	57.8	57.9	94.3	3.2	39.3	**100.0**	96.4	32.1	23.6	87.9	43.3
DCL	**78.2**	70.7	71.1	79.7	91.8	0.8	78.5	**100.0**	99.3	24.8	**57.9**	**100.0**	**78.1**
Ours	**78.2**	**73.4**	**74.3**	**95.1**	88.7	4.8	84.0	96.0	97.1	**73.8**	28.4	**100.0**	75.0

the feature extraction layer of ResNet50, making the parameters lower from 3.58×10^7 to 6.74×10^6 and 8.06×10^5, respectively. As shown in Table V, although the parameters of the student model are reduced by one and two orders of magnitude, it still outperforms the teacher model. Since the proposed weakly correlated distillation suppresses feature noises generated by teachers.

C. Comparison With SOTA

We compare the proposed method with five SOTA methods, including progressive multigranularity training (PMG) [53], reducing style bias (Sagnet) [54], routing diverse distribution (RIDE) [55], diversity consistency learning (DCL) [10], and TLC [9]. We perform the experiment according to the model parameters set in the author's article.

1) Accuracy: We evaluate the recognition accuracy of our method and the SOTA methods by testing on DOTA dataset. As shown in Table VI, ours achieves an accuracy of 91.2%, the highest accuracy compared with SOTA methods. Specifically, our method improves PMG, TLC, Sagnet, RIDE, and DCL by 20.3%, 12.2%, 9.7% 8.8%, and 1.3%, respectively.

2) Generalization: We further verify the generalization of our method and SOTA methods on HRRSD and NWPU VHR-10 datasets. As can be seen from Tables VII and VIII, our method achieves the best performance on HRRSD and NWPU VHR-10 datasets, whose accuracy is 63.0% and 78.2%, precision is 71.2% and 73.4% and recall is 63.7% and 74.3%, respectively, outperforming the second-best method by 5.4% and 2.0% on average.

Our method achieves superior results in the comparisons to some other methods since two strategies are implemented: 1) teacher models are built on large-scale natural image datasets for acquiring prominent feature extraction ability and 2) weakly correlated distillation selects the weakly correlated features to distill the student that suppresses feature noises generated by teachers.

3) Visualization: The Table represents the probability of being predicted as the correct category, and the corresponding confusion matrix further represents the probability of being predicted as each category. Tables and confusion matrices are one-to-one correspondence (C.), i.e., Table VI C. Fig. 6, Table VII C. Fig. 7, Table VIII C. Fig. 8. Confusion matrices of

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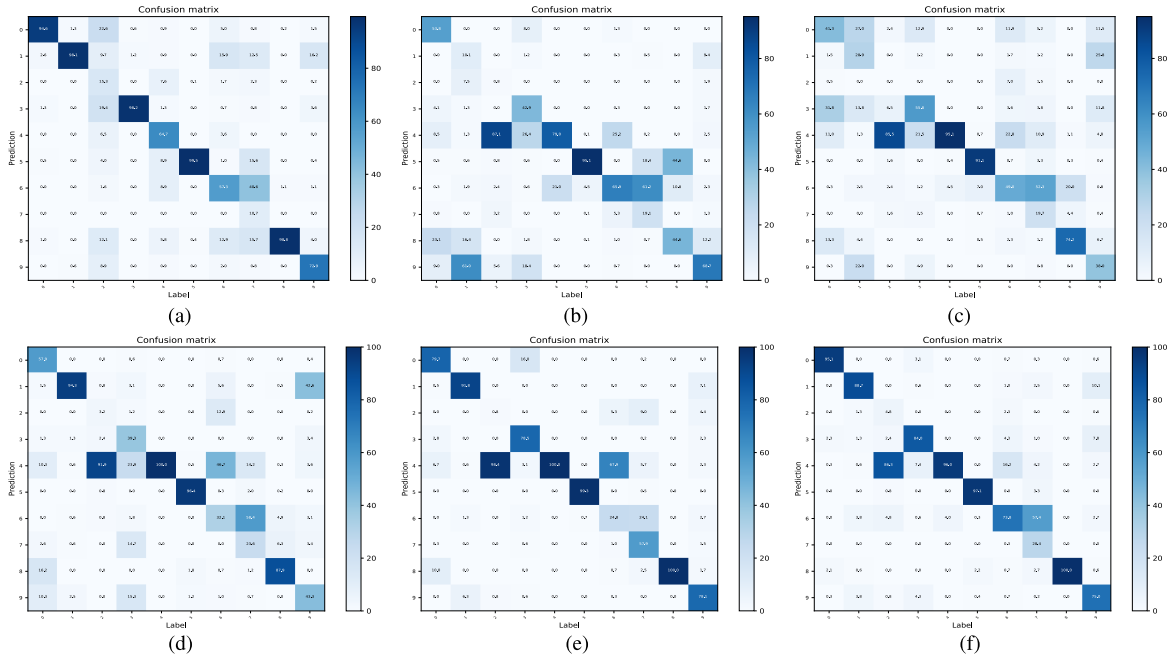


Fig. 8. Confusion matrices of different methods on NWPU VHR-10. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. (a) PMG. (b) TLC. (c) Sagnet. (d) RIDE. (e) DCL. (f) Ours.

different methods on DOTA are shown in Fig. 6. The abscissa in the figure represents the real categories and the ordinate represents the predicted category. Each column represents the prediction result for the current category, and the sum of the elements is approximately equal to 1 due to the abandonment of decimal places. In Fig. 6, our method has a better overall effect than other comparison methods. In addition, we visualize the results of verifying generalization on HRRSD and NWPU VHR-10, as shown in Figs. 7 and 8. Our method has relatively higher prediction results on the NWPU VHR-10 dataset, indicating the generalization of our method.

V. Conclusion

We make the attempt to study the remote sensing object recognition with limited labels and present the weakly correlated distillation learning model. Firstly, we design multi-teacher distillation model which stored information trained by large labeled datasets and fine-tune them with a limited labeled remote sensing dataset. However, because teacher models are trained with limited labeled remote-sensing dataset, they will generate noise. Therefore, in order to offset the noise in teacher models, we use weakly correlated distillation to select appropriate teacher models to distill the student model. Extensive experiments are built on three widely-used datasets of DOTA, HRRSD and NWPU VHR-10 that verify the superior performance of our method. Possible future directions are small-amount-of-sample low-resolution and fine-grained remote sensing object recognition.

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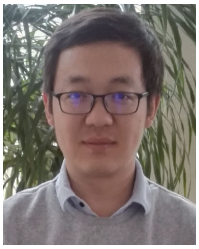
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